

Release Outputs Required

Required final outputs

Required Output	Owner	Release Criteria	Current State
Signed mechanical specification	Mechanical / system engineering	295.0 x 140.0 mm envelope, 56.0 mm height note, datum, rails, keepouts, holes, connector exits, heatsink and airflow confirmed.	Reference envelope modeled; signoff pending.
Native schematic and schematic PDF/SVG	Electrical engineering	Dual NCE, TFLN PIC, RF/bias, HBM, PCIe/control/power blocks visible and aligned to netlist.	Included in schematic reference; parity closure pending.
Final netlists	Electrical engineering	Generated from closed schematic and loaded into PCB with no unexplained differences.	Reference netlists included; final netlist after ECO closure.
PCB native project	PCB layout engineering	Updated KiCad project with DRC 0 and no unintended unconnected items.	Current project included in ECO package; DRC/parity still open.
Gerbers, drill, drill map, stackup, impedance notes	PCB layout / manufacturer CAM	Regenerated from DRC-clean PCB and approved by DFM.	Review outputs exist; regenerate after ECO closure.
Placement and assembly outputs	PCB layout / assembly engineering	XY placement, centroid, assembly drawings, polarity and DNP markings match final BOM.	Release-pending pending BOM and parity closure.
Final procurement BOM	Component / procurement engineering	MPN, value, package, quantity, source/AVL, DNP, alternates, revision/date signed.	Current BOM is release-pending; final BOM must follow ECO closure.
Validation report	Electrical / PCB verification	DRC 0, unconnected 0, schematic parity closed or signed-waived, ERC reviewed.	DRC 32, unconnected 0, schematic parity 564.
DFM/DFA closure report	Manufacturer with engineering review	CAM, panelization, solder mask, silkscreen, drill, stackup, assembly and test risks closed.	Pending ECO-clean outputs.
Final release manifest	Release engineering	All final files listed with SHA256, revision, date, and owner approval.	This do-over package includes its own manifest; production manifest comes after closure.